Board Policy 3.30.080-P: PPS Climate Crisis Response, Climate Justice and Sustainable Practices Policy

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Summary of Revisions

In the introduction:

> *Current language:*
> Our goal is for all students to be environmentally literate, and to understand the connections between their daily actions and the natural world ~~by using community-based learning and civic action as a context for meeting academic goals described in the Graduate Portrait~~.
>
> *Revise to:*
> Our goal is for all students to be environmentally literate, and to understand the connections between their daily actions and the natural world, as well as the larger social, economic, and infrastructural systems that drive environmental change, climate disruption, and public health. The District supports students' environmental literacy by using community-based learning and civic action as a context for meeting academic goals described in the Graduate Portrait.

In Pillar 1, under Goal 1.3:

> *Current language:*
> 1. Increase shade trees, green spaces, and de-pave projects, with highest priority in low-canopy neighborhoods.
>
> *Revise to:*
> 1. Increase shade trees, green spaces, and de-pave projects, with highest priority in low-canopy neighborhoods and Portland's urban heat islands.

In Pillar 2: Improve Health and Wellness, Goal 2.1:

> *Current language:*
> **Goal 2.1: PPS will address climate-based impacts on health, safety, and wellness of its students and employees.**

1. Include climate change impacts, such as flooding, landslides, and wildfires, as risks in district real property asset assessment and management.
2. Where possible, prioritize foods that are local, organic, seasonal, plant-based, minimally processed and/or minimally packaged, while meeting requirements of the USDA Child Nutrition Programs.

Revise to:

Goal 2.1: PPS will address environmental health and climate-based impacts on health, safety, and wellness of its students and employees.

1. Include climate change impacts, such as flooding, landslides, and wildfires, as risks in district real property asset assessment and management.
2. Establish and regularly update emergency response plans for climate-related extreme events to prevent foreseeable facility damage and protect student and staff safety. Develop public plans for ice storms and snow, wildfire smoke, and flooding; maintain and improve current heat illness/extreme heat plans. Protocols should include pre-event building preparation (such as shutting off water mains and preparing HVAC systems for extreme conditions) and clear communication with families and staff.
3. Address excessive heat and cold in buildings through measures such as installing new HVAC systems, restoring operable windows that were sealed up in past decades, optimizing window restrictors to balance safety with effective heat relief, adjusting window-use policies to allow overnight cooling during heat events while maintaining security, expanding shade structures and trees, deploying temporary heating and cooling systems, and other adaptive strategies that protect health and learning. Generators for heating and cooling systems should be electric and avoid fossil fuels to the maximum extent practicable.
4. To protect health, equity, and learning, continuously improve indoor air quality by pairing ventilation and in-room filtration. Use energy-efficient ways to increase ventilation to a goal of 6 effective air changes per hour; use quiet, effective in-room filtration strategies to provide more clean air and increase effective air changes per hour beyond 6 and closer to 12. Do not purchase technologies marketed for air cleaning that lack the scientific evidence for safety and effectiveness that ventilation and filtration have. Examples include electronic air cleaning technologies and chemical room-fogging systems marketed as air cleaners, whether stand-alone or for installation in HVAC systems. Issue guidance to turn these features off when present in air purifiers.
5. Transition air quality monitoring from a complaint-based system to installation of continuous, real-time, in-room sensors that provide publicly-available readings, including carbon dioxide, temperature, humidity, and $PM_{2.5}$. Interim targets for parameters: <1000 ppm carbon dioxide, >60 degrees Fahrenheit, heat index <80 degrees Fahrenheit, $PM_{2.5}$ under 5 $\mu g/m^3$; long-term goals: <750 ppm carbon dioxide, 68-74 degrees Fahrenheit, strive for measuring airflows every 5 years. Use monitoring results as an opportunity for education and communication with staff, students, and community to deepen understanding of how indoor air quality

impacts transmission of respiratory infections, asthma and allergy incidence, cardiopulmonary health, cognitive performance, and student learning and attendance; support timely action when parameters fall outside targets; and reinforce practices that maintain healthy classroom environments.

6. Where possible, prioritize foods that are local, organic, seasonal, plant-based, minimally processed and/or minimally packaged, while meeting requirements of the USDA Child Nutrition Programs.

Under Glossary of Definitions, add the following definitions:

Airflows: Volumes of outdoor air, total air, and air recirculated through MERV-rated filters per unit time delivered to each room via the HVAC system measured via test, adjust, balance methods, and volumes of clean air per unit time delivered by in-room air filtration. Typically in cubic feet per minute, which can also be converted to air changes per hour or per-person rates.

Effective air changes per hour: A measure of how many times in one hour ventilation and filtration systems deliver a volume of air equal to the size of the room. Effective air changes per hour include outdoor air from ventilation, recirculated air filtered through MERV-rated filters in HVAC systems, and in-room filtered air.

Electronic air cleaning technologies: Devices or features that use electricity, electric fields, UV light, or other additive technologies to produce ions, charged airborne particles, activated metal catalysts, or other reactive compounds. Examples include bipolar ionization, corona discharge, electrostatic precipitators, ionizers, needle point discharge, ozone, photocatalytic oxidation, plasma, and reactive oxygen species. These technologies can unintentionally alter room chemistry and produce harmful contaminants, such as ultrafine particles, formaldehyde, and other volatile organic compounds.

In-room filtration: The process of moving air through mechanical fibrous filters in air purifiers located inside classrooms and other occupied school spaces in order to remove airborne particles and aerosols. Air purifiers that can provide in-room filtration have a measurable clean air delivery rate (the volume of clean air the device supplies per unit time) and include portable floor models, wall-mounted filters, table-top units, and furniture- or ceiling-integrated units. These devices may operate via electronic / digital control panels needing active management (daily turning on by staff or students) or with mechanical switch-based designs that can be paired with external timers or motion sensors to enable automatic operation.

$PM_{2.5}$: Fine particulate matter in air; a mix of inhalable solid particles and liquid droplets with diameters of 2.5 micrometers or smaller. Sources of PM2.5 include vehicle exhaust, wildfires, wood burning for winter home heating, dust, mold, combustion from school boilers and heating equipment, cooking operations in school kitchens, and secondary formation from indoor chemical reactions.

Urban heat islands: Areas within cities that lack adequate vegetation and become significantly hotter due to concrete and asphalt absorbing, retaining, and re-releasing heat.

(See also: EPA EnviroAtlas Urban Heat Islands and Vulnerable Communities in Portland, OR)

Ventilation: The process of supplying and removing air within a building to maintain indoor air quality; includes both outdoor air brought in through mechanical or natural ventilation systems and indoor air recirculated through MERV 13 or higher-rated filters to remove respiratory aerosols, particulates, and other filterable air pollutants.

Revisions in Context of Full Policy

In response to the human-caused climate crisis currently underway and the direct harm being done to our District, society, and planet, Portland Public Schools (PPS) is committing to immediately mobilize resources for climate action. To this end, the District commits to reducing greenhouse gas (GHG) emissions and minimizing other negative environmental impacts; improving our school communities' health and wellness; and building a culture of learning, responsibility, and sustainability centered on our values of racial equity and climate justice.

Climate change disproportionately impacts the vulnerable members of our community. Implementation of this policy will prioritize serving people with disabilities, communities of color, and other vulnerable populations.

The District aims to educate and empower students as leaders in the just transition to a sustainable city and restorative society. Our goal is for all students to be environmentally literate, and to understand the connections between their daily actions and the natural world, as well as the larger social, economic, and infrastructural systems that drive environmental change, climate disruption, and public health. The District supports students' environmental literacy by using community-based learning and civic action as a context for meeting academic goals described in the Graduate Portrait.

The District shall prioritize investments that achieve the greatest emissions reduction, while integrating climate action into our curriculum and developing a culture of environmental stewardship and climate justice throughout our organization. Other efforts which have a lower impact on emissions and require major changes in infrastructure can be implemented over the longer term.

Our two overarching objectives are:

> **Emissions Reduction:** PPS will reduce its greenhouse gas emissions by 50 percent by 2030, using the 2018-2019 school year baseline, and reach net zero emissions by 2040.
>
> For the sake of emissions tracking, PPS will use a Scope 1 and Scope 2 analysis, which includes all emissions associated with the following: district-operated buses and fleet vehicles, and building energy use from both natural gas and electricity. In addition to Scope 1 and Scope 2, PPS will also track emissions associated with waste disposal (food waste, recycling, and landfill-bound garbage), which is defined as part of a Scope 3 analysis.
>
> **Engagement, Resilience, and Wellness:** The District will take steps to prepare schools for the effects of climate change, and will ensure staff and students learn about and engage in climate solutions, climate resiliency, and climate justice practices.

As the District works to meet these climate and sustainability goals and targets, it will maximize, minimize, and take other actions to reduce carbon emissions and increase sustainable practices when alternatives exist that meet District standards, are aligned with carbon-reduction targets, are cost competitive over time, and available technology exists.

It should be noted that these goals are based on the following assumptions: technology in the electric vehicle market will advance significantly; the cost of electric school buses will come down significantly over the next 10-20 years; the District will continue to pass voter-approved bonds on a regular cycle to support infrastructure investment; and the District will allocate general fund budget to help reach these goals.

We will align our work with the U.S. Department of Education's Green Ribbon School Pillars. In accordance with those pillars, PPS is adopting the following goals:

Pillar 1: Reduce Environmental Impact and Costs
Goal 1.1: PPS will design and construct new low-carbon schools and renovations that are energy-efficient, resilient, and adaptable.

1. Use appropriate industry standards (e.g. LEED Gold, Net-Zero ready, Oregon's Energy Ready Commercial Code) when designing new and modernized buildings.
2. Prohibit the installation of fossil fuel infrastructure (gas-fired equipment) in all new buildings.
3. Phase out fossil fuel infrastructure (gas-fired equipment) in all existing buildings by 2050.
4. Increase energy efficiency, maximize the use of renewable energy sources, and minimize the use of fossil fuels.
5. Limit the amount of refrigerants used. Manage existing refrigerants with best available technology.
6. Transition to building materials produced with less carbon and that are more sustainable.
7. Design, renovate, and construct new facilities to improve resiliency to climate-related emergencies.

Goal 1.2: PPS will maximize reductions in Green House Gas emissions from district operations, maintenance, and facilities management.

1. Maximize efficiency in fuel, electricity, and water use.
2. Establish standardized systems for waste prevention practices, including recycling and food waste reduction at campuses.
3. Minimize disposable materials.

Goal 1.3: PPS will maximize the carbon sequestration potential and other environmental benefits of green school yards and increase the ability of school grounds to adapt to climate extremes.

1. Increase shade trees, green spaces, and de-pave projects, with highest priority in low-canopy neighborhoods and Portland's urban heat islands.
2. Maximize onsite stormwater management in compliance with city code.
3. Establish green school yard plans to care for school grounds.

4. Phase out gas-powered equipment used for grounds maintenance.

Goal 1.4: PPS will minimize greenhouse gas emissions from student and staff transportation, including transitioning to electric or low-emission vehicles.

1. Establish school-based infrastructure and support for staff and students to walk, bike, bus or telecommute, in order to reduce single-occupant car commuting.
2. Transition PPS and contracted vehicle fleet to electric or low emissions or renewable fuels. Work with industry, utilities, and the state and federal governments to reduce the costs of shifting towards electric or low-emission vehicles.
3. Educate and communicate to students, parents and staff about benefits of and opportunities to reduce single-occupant car commuting and idling.

Goal 1.5: PPS will reduce the demand for new materials and resources, and procure materials, products, and services in a manner that integrates climate considerations, fiscal responsibility, and equity priorities.

1. Where applicable, use best practices related to Environmentally Preferable Purchasing for centralized, formally solicited purchases, and provide tools and guidance to schools to support sustainable, climate-preferred product and services purchasing.
2. Transition to increased electronic communication and reduce paper copying and maximize recycled content in paper used.
3. PPS will produce a report at least annually of PPS investments in fossil fuels with information that, in light of state-mandated investments, is reasonably accessible and available to it. Investments in state investment funds will be reported, recognizing that the Oregon State Treasury controls the decision-making for state investment funds.

Pillar 2: Improve Health and Wellness

Goal 2.1: PPS will address environmental health and climate-based impacts on health, safety, and wellness of its students and employees.

1. Include climate change impacts, such as flooding, landslides, and wildfires, as risks in district real property asset assessment and management.
2. Establish and regularly update emergency response plans for climate-related extreme events to prevent foreseeable facility damage and protect student and staff safety. Develop public plans for ice storms and snow, wildfire smoke, and flooding; maintain and improve current heat illness/extreme heat plans. Protocols should include pre-event building preparation (such as shutting off water mains and preparing HVAC systems for extreme conditions) and clear communication with families and staff.
3. Address excessive heat and cold in buildings through measures such as installing new HVAC systems, restoring operable windows that were sealed up in past decades, optimizing window restrictors to balance safety with effective heat relief, adjusting window-use policies to allow overnight cooling during heat events while maintaining security, expanding shade structures and trees, deploying temporary heating and cooling systems, and other adaptive strategies that protect health and learning. Generators for heating and cooling systems should be electric and avoid fossil fuels to the maximum extent practicable.

4. To protect health, equity, and learning, continuously improve indoor air quality by pairing ventilation and in-room filtration. Use energy-efficient ways to increase ventilation to a goal of 6 effective air changes per hour; use quiet, effective in-room filtration strategies to provide more clean air and increase effective air changes per hour beyond 6 and closer to 12. Do not purchase technologies marketed for air cleaning that lack the scientific evidence for safety and effectiveness that ventilation and filtration have. Examples include electronic air cleaning technologies and chemical room-fogging systems marketed as air cleaners, whether stand-alone or for installation in HVAC systems. Issue guidance to turn these features off when present in air purifiers.
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6. Where possible, prioritize foods that are local, organic, seasonal, plant-based, minimally processed and/or minimally packaged, while meeting requirements of the USDA Child Nutrition Programs.

Goal 2.2: PPS will support frontline student communities to build resilience from climate change induced stresses and support preparation for and recovery from these events.

1. PPS will coordinate with state, regional, and local jurisdictions and community-based organizations to communicate with and support PPS students and families during climate-related events and natural disasters, such as wildfires, flooding, drought, heat waves, extreme winter storms, extreme wind events. The District will make available its facilities, as appropriate, during community response efforts to climate events.
2. Incorporate climate justice priorities and climate resiliency design to inform long-term facility planning. Plans should prioritize serving people with disabilities and frontline communities.

Pillar 3: Provide Effective Environmental and Sustainability Education

Goal 3.1: PPS will empower staff as allies for a healthy climate.

1. Provide learning opportunities for staff on climate science/climate justice.
2. Continuously collaborate with staff to utilize sustainability practices in the performance of their work and provide information, support, and tools to help implement those practices.

Goal 3.2: With guidance from frontline students and communities, PPS will develop curricular learning opportunities, so PPS graduates know the causes and consequences

of climate change, understand climate justice, and have opportunities to practice climate solutions.

1. Develop and deliver curriculum and resources to help students understand, prepare for, and respond to climate change impacts.
2. Teaching and learning about climate change will recognize and support the emotional and other impacts of climate change on individuals.
3. Provide opportunities for students to probe the underlying causes of the climate crisis and the structural racism embedded in climate change due to actions by majority white countries with disproportionate negative burdens and impacts on people of color, and to understand the dynamic nature of complex systems and how they change, and opportunities for them to practice using systems thinking to inform their choices.

Goal 3.3: PPS staff will collaborate with students to create opportunities to engage youth in hands-on climate learning, preparation, and practice on a regular basis at all PPS schools.

1. Support development of youth leadership and engagement opportunities around climate solutions, prioritizing opportunities for students from frontline communities.
2. Support student-led climate justice, climate action, and climate resiliency initiatives.
3. Treat the grounds and garden of each school as a learning space and involve students in nature-based grounds improvements and design. Support opportunities for students in every PPS elementary school to learn in a designated outdoor learning space.
4. Support students in their ongoing and critical role as climate response, climate justice, and sustainability advocates.

CLIMATE CRISIS RESPONSE COMMITTEE

To monitor effective implementation, transparency, and tracking of progress, PPS will establish a committee of no more than nine persons that is made up of a majority of members who identify as people of color, and two members who are current PPS students. The Board shall approve the Charter for and members of the Committee. The Committee will convene no less than quarterly and will report to the Board on an annual basis the progress the District is making under the policy directives. The Committee informs and advises the Board but does not have decision-making authority.

Glossary of Definitions

Scope 1 Emissions: direct greenhouse (GHG) emissions that occur from sources that are controlled or owned by an organization (e.g., emissions associated with fuel combustion in boilers, furnaces, water heaters, vehicles).

Scope 2 Emissions: indirect GHG emissions associated with the purchase of electricity, steam, heat, or cooling. Although scope 2 emissions physically occur at the facility where they are generated, they are accounted for in an organization's GHG inventory because they are a result of the organization's energy use.

Scope 3 Emissions: the result of activities from assets not owned or controlled by the reporting organization, but that the organization indirectly impacts in its value chain. Scope 3 emissions include all sources not within an organization's scope 1 and 2 boundary. The scope 3 emissions for one organization are the scope 1 and 2 emissions of another organization. Scope 3 emission sources include emissions both upstream and downstream of the organization's activities.

(Scope emissions defined by EPA Center for Corporate Climate Leadership Home)

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Feasible: capable of being done or carried out. Feasibility will include the following factors: economics/available funding and appropriate trade-offs (i.e. the district will not spend general funds on energy efficiency if it is in lieu of incremental staffing in schools).

Frontline Communities: Communities that experience continuing injustice—including people of color, immigrants, people with lower incomes, those in rural areas, and indigenous people—and face a legacy of systemic, largely racialized, inequity that influences their living and working places, the quality of their air and water, and their economic opportunities.

In-room filtration: The process of moving air through mechanical fibrous filters in air purifiers located inside classrooms and other occupied school spaces in order to remove airborne particles and aerosols. Air purifiers that can provide in-room filtration have a measurable clean air delivery rate (the volume of clean air the device supplies per unit time) and include portable floor models, wall-mounted filters, table-top units, and furniture- or ceiling-integrated units. These devices may operate via electronic / digital control panels needing active management (daily turning on by staff or students) or with mechanical switch-based designs that can be paired with external timers or motion sensors to enable automatic operation.

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Systems Thinking: a holistic approach to analysis that focuses on the way that a system's constituent parts interrelate and how systems work over time.

Urban heat islands: Areas within cities that lack adequate vegetation and become significantly hotter due to concrete and asphalt absorbing, retaining, and re-releasing heat.

(See also: EPA EnviroAtlas Urban Heat Islands and Vulnerable Communities in Portland, OR)

Ventilation: The process of supplying and removing air within a building to maintain indoor air quality; includes both outdoor air brought in through mechanical or natural ventilation systems and indoor air recirculated through MERV 13 or higher-rated filters to remove respiratory aerosols, particulates, and other filterable air pollutants.